

I- Choose the correct answer (1.25 pts/correct answer)

- 1- In the graphical method for solving linear programming problems, the optimal solution lies:
- a- Anywhere inside the feasible region
 - b- At the intersection of constraint lines
 - ☒ c- At a corner (extreme) point of the feasible region
 - d- At the midpoint of the objective function line
- 2- The solution of a transportation problem with 6 rows (sources) and 5 columns (destinations) is feasible if the number of basic variables is:
- a- 9
 - ☒ b- 10
 - c- 11
 - d- 12

- 3- What is the feasible solution in a graphical method-based problem with the following constraints:

$$X + Y > 10$$

$$2X + Y < 5$$

$$X + 3Y < 8$$

- ☒ a- No feasible solution.
 - b- The point with $X=5$ and $Y=5$.
 - c- The line between two points $A(5,5)$ and $B(5,8)$.
 - d- None of the above.
- 4- A graphical method is not suitable when:
- a- The number of constraints is greater than 2.
 - ☒ b- The number of variables is greater than 2.
 - c- The problem is a maximization problem.
 - d- None of the above.

I- Treat ONLY three of the following questions: (5 pts/question):

- 1- Given that the variable x represents the number of the cars produced by day, the variable y represents the number of bikes produced per day and Z represents the income in (\$)

According to the following constraints :

$$x + y \leq 500$$

$$x \leq 2y$$

$$y \geq 100$$

$$x, y \geq 0$$

Solve the problem using the graphical method, knowing that $\text{Max } Z = 5x + 3y$

- 2- Maximize the following economical function:

$$Z = 3X_1 + 2X_2 + X_3$$

Under the constraints:

$$X_1 - X_2 + X_3 = 15$$

$$3X_1 + X_2 \geq 10$$

$$X_2 + X_3 \leq 10$$

$$X_1, X_2, X_3 \geq 0$$

- 3- Given the following transportation problem showing the costs between sources and destinations:

From \ To	Saida	Zahle	Chtoura	Available Quantities
Dekwaneh	2	1	3	10
Hamra	4	5	6	20
Demanded Quantities	15	6	4	

- a- Find an initial basic solution by using Vogel's Approximation Method (VAM).
- b- Test if the obtained basic solution is optimal or not.

- 4- The following table gives the activities for elaboration of a project P:

Activity	Immediate Predecessor(s)	Duration in Days
A	-	10
B	3 days after the starting date of the project	21
C	-	8
D	A, B	5
E	2/3 B	15
F	B, 5/8 C	6
G	D, E, F	4
H	C, E	10

Construct the PERT network of this project.

I)

$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
1	2	3	4
c	b	a	b

OR
P1/5

Accepted Area

$\frac{1}{2}$ each line $\times 3 = 1\frac{1}{2}$ pts
 $\frac{1}{4}$ Draw

1) Max $Z = 5x + 3y$

$$x + y \leq 500$$

$$x + y \leq 500 \Rightarrow$$

$$(1) \begin{array}{c|c|c} x & y & \\ \hline 0 & 500 & 500 \\ 500 & 0 & 0 \end{array}$$

$$x \leq 2y \Rightarrow x - 2y \leq 0$$

$$A(0, 500) \quad B(500, 0)$$

$$y \geq 100 \Rightarrow y \geq 100$$

$$(2) \begin{array}{c|c|c} x & y & \\ \hline 0 & 200 & 200 \\ 200 & 100 & 100 \end{array}$$

$$x, y \geq 0 \quad x, y \geq 0$$

$$C(200, 100)$$

$$D \in (3) \Rightarrow y_D = 100$$

$$D \in y'oy \Rightarrow x_D = 0 \quad D(0, 100)$$

$$(3) y = 100$$

$$E = (1) \cap (2)$$

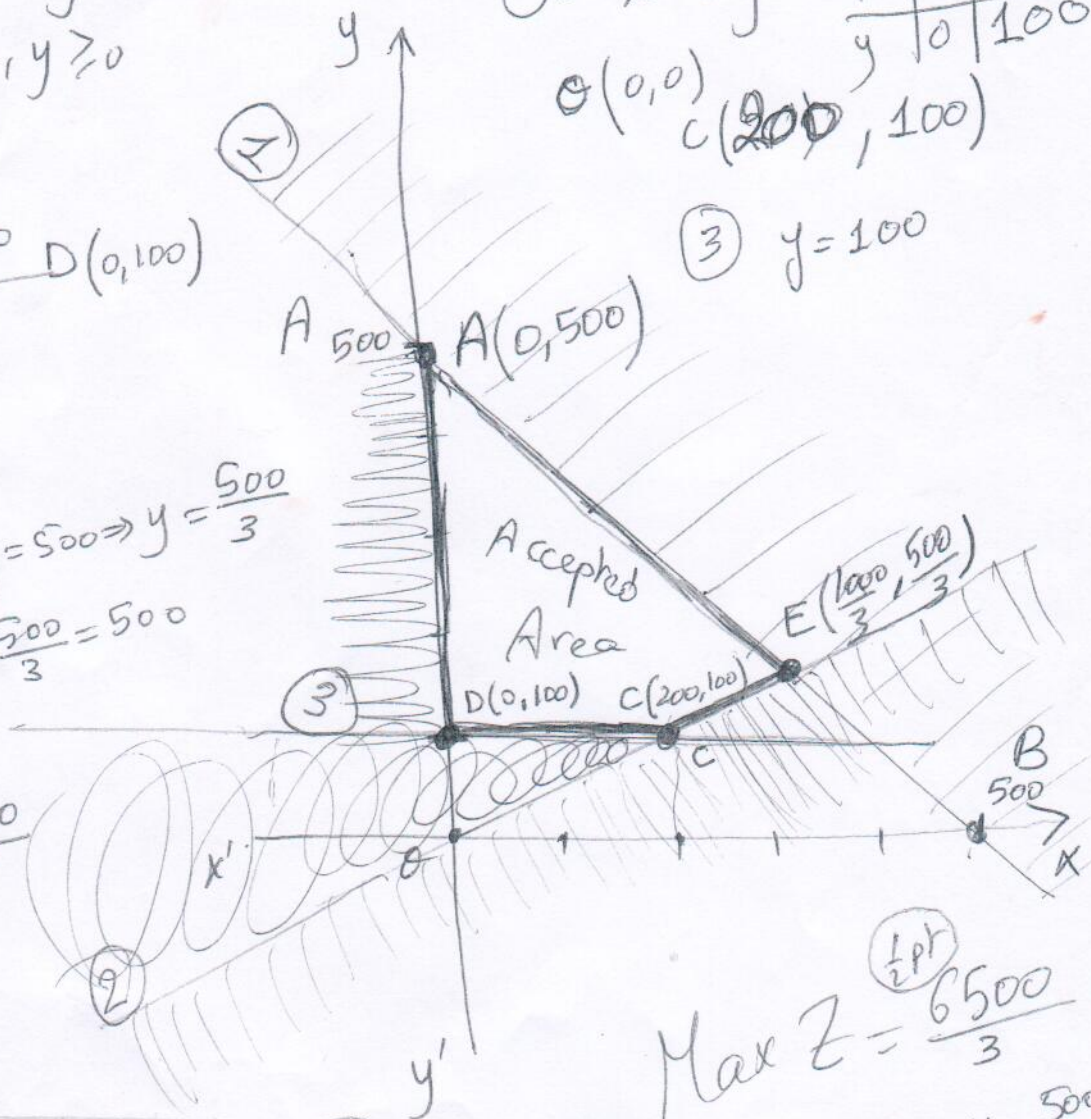
$$\begin{cases} x + y = 500 \\ x - 2y = 0 \end{cases}$$

$$y + 2y = 500 \Rightarrow 3y = 500 \Rightarrow y = \frac{500}{3}$$

$$x + y = 500 \Rightarrow x + \frac{500}{3} = 500$$

$$x = 500 - \frac{500}{3} = \frac{1500 - 500}{3} = \frac{1000}{3}$$

$$E\left(\frac{1000}{3}, \frac{500}{3}\right)$$



P5 $\frac{1}{4} \times 4 = 1$ pt $Z = 5x + 3y$ $\frac{1}{4} \times 4 = 1$ pt

$$D(0, 100) \quad Z_D = 5(0) + 3(100) = 300$$

$$C(200, 100) \quad Z_C = 5(200) + 3(100) = 1300$$

$$E\left(\frac{1000}{3}, \frac{500}{3}\right) \quad Z_E = 5\left(\frac{1000}{3}\right) + 3\left(\frac{500}{3}\right) = \frac{5000}{3} + \frac{1500}{3} = \frac{6500}{3}$$

$$A(0, 500) \quad Z_A = 5(0) + 3(500) = 1500$$

$$\text{Max } Z = \frac{6500}{3}$$

$$\Rightarrow x = \frac{1000}{3}, y = \frac{500}{3}$$

$$\frac{1500}{3} = \frac{6500}{3}$$

$$x_1 + x_2 + x_3 = 15 \Rightarrow x_1 - x_2 + x_3 + a_1 = 15$$

$$3x_1 + x_2 \geq 10 \Rightarrow 3x_1 + x_2 - e_1 + a_2 = 10$$

$$x_2 + x_3 \leq 10 \Rightarrow x_2 + x_3 + e_2 = 10$$

$$x_1, x_2, x_3 \geq 0$$

$$(3eq + 7) = 4eq \times \frac{1}{4} = 1pt$$

m Constant $\rightarrow +\infty$

$$\text{Max } Z = 3x_1 + 2x_2 + x_3 \Rightarrow$$

$$= 3x_1 + 2x_2 + x_3 - ma_1 - ma_2$$

$$= 3x_1 + 2x_2 + x_3 - m(a_1 + a_2)$$

$$= 3x_1 + 2x_2 + x_3 - m(15 - x_1 + x_2 - x_3 + 10 - 3x_1 - x_2 + e_1)$$

$$Z = 3x_1 + 2x_2 + x_3 - m(25 - 4x_1 - x_3 + e_1)$$

$$Z = 3x_1 + 4mx_1 + 2x_2 + x_3 + mx_3 - me_1 - 25m$$

$$Z + (-4m-3)x_1 - 2x_2 + (-m-1)x_3 + me_1 = -25m$$

	x_1	x_2	x_3	a_1	a_2	e_1	e_2	Sol	R
a_1	1	-1	1	1	0	0	0	15	15
a_2	3	1	0	0	1	-1	0	10	10/3
e_2	0	1	1	0	0	0	1	10	∞
Z	-4m-3	-2	-m-1	0	0	m	0	-25m	
a_1	0	-4/3	1	1		1/3	0	35/3	35/3
x_1	1	1/3	0	0		-1/3	0	10/3	∞
e_2	0	1	1	0		0	1	10	10
Z	0	4/3 m - 1	-m - 1	0		-m/3 - 1	0	-35/3 m + 10	

3x First 4 Table
= 3pts
Cal $\frac{1}{4}x + \frac{1}{4}P.C. + \frac{1}{4}$

Last Table 4pt

	x_1	x_2	x_3	a_i	e_1	e_2	Sol	R	OR (P3)/5
a_1	0	$-\frac{7}{3}$	0	1	$\frac{1}{3}$	-1	$\frac{5}{3}$	5	
x_1	1	$\frac{1}{3}$	0	0	$-\frac{1}{3}$	0	$\frac{10}{3}$	-10	
x_3	0	1	1	0	0	1	10	∞	
Z	0	$\frac{7}{3}m$	0	0	$-\frac{m}{3}$	$m+1$	$-\frac{5}{3}m$	+20	
e_1	0	-7	0		1	-3	5	-5/7	
x_1	1	-2	0		0	-1	5	-5/2	
x_3	0	1	1		0	1	10	10	
Z	0	-7	0		0	-2	25		
e_1	0	0	7		1	4	75		
x_1	1	0	2		0	1	25		
x_2	0	1	1		0	1	10		
Z	0	0	7		0	5	95		

All non basic variables in Z are $\geq 0 \Rightarrow$ optimal sol
 Reached Max Z = 95, $x_1 = 25$, $x_2 = 10$, $x_3 = 1$
 $\frac{1}{4}$ pt $\frac{1}{2}$ pt

P3/5

№3

2 1 3 10

4 5 6 20

15 6 4 $\begin{matrix} 30 \\ 25 \end{matrix}$ $\left(\frac{3}{2}\right)$

OR
P4/5

Add Dummy Column $\left(\frac{1}{2}\right)$

5 Basic var $\times \frac{1}{4}$
= $\left(1\frac{1}{4}\right)$

20	$\textcircled{1}$	$\textcircled{3}$	0	10/4/2	1	1	1
4/15	5	6	$\textcircled{4}$	20/15	$\textcircled{4}$	1	2
15	6	4	5				
	0	0	0				

2	4	3	0
2	$\textcircled{4}$	3	-
2	-	$\textcircled{3}$	-

basic sol:

0) $4 = 2 + 4 - 1 = 5 \checkmark$ $Z = 6 + 12 + 60 + 0 = 78$ $\left(\frac{1}{2}\right)$

0) $V_1=2$ $V_2=1$ $V_3=3$ $V_4=-2$

	2	1	3	0
$U_1=0$	0	6	4	
$U_2=2$	15	5	6	5

$\bar{C}_{14} = -2$
 $\bar{C}_{22} = -2$
 $\bar{C}_{23} = -1$

u, v $\left(\frac{1}{2}\right)$
 $\bar{C}$ $\left(\frac{1}{2}\right)$

All $\bar{C}_{ij} \leq 0 \Rightarrow$ optimal solution $\left(\frac{1}{4}\right)$

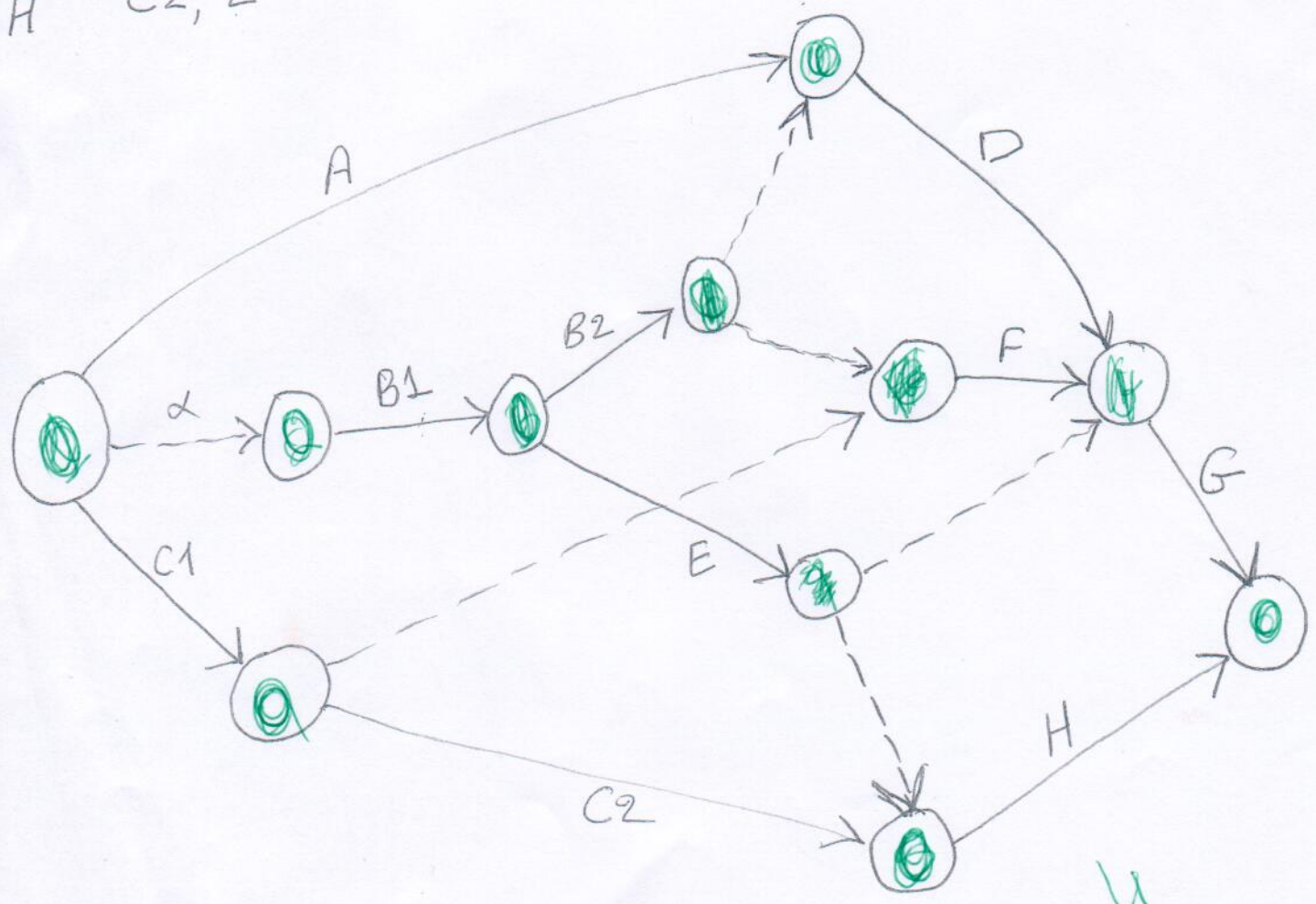
~~Max Z = 78~~

II No. 4

A	-	10
α	-	3
B1	α	14
B2	B1	7
C1	-	5
C2	C1	3
D	A, B2	5
E	B1	15
F	B2, C1	6
G	D, E, F	4
H	C2, E	10

OR
P 5/5

$\frac{1}{2}$ each Task (~~or D~~) $\times 10$ max $\Rightarrow$ 5pts



P 5/5

u

$\frac{1}{4} \times 4 = (5 \text{ pts})$

II 1- Max $Z = 5x + 3y$

① $x + y \leq 500$ A(0,500) B(500,0)

② $x - 2y \leq 0$ O(0,0) C(200,100)

③ $y \geq 100$ $y = 100$

pts $\frac{4 \times 1}{4} = 1 \text{ pt}$

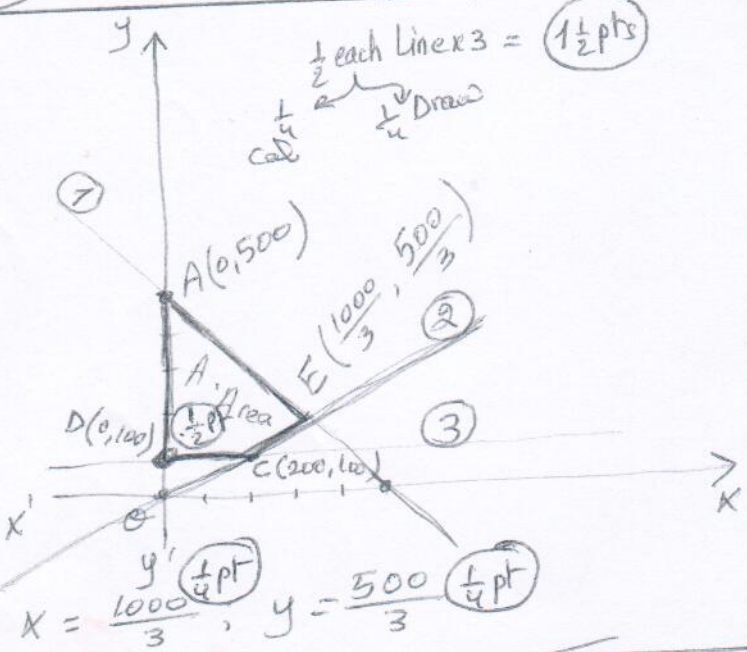
D(0,100) $Z_D = 300$

C(200,100) $Z_C = 1300$

E($\frac{1000}{3}, \frac{500}{3}$) $Z_E = \frac{6500}{3}$

A(0,500) $Z_A = 1500$

Max $Z = \frac{6500}{3}$



2 $x_1 - x_2 + x_3 = 15 \Rightarrow x_1 - x_2 + x_3 + a_1 = 15$

$3x_1 + x_2 \geq 10 \Rightarrow 3x_1 + x_2 - e_1 + a_2 = 10$

$x_2 + x_3 \leq 10 \Rightarrow x_2 + x_3 + e_2 = 10$

Max $Z = 3x_1 + 2x_2 + x_3 \Rightarrow \text{Max } Z = 3x_1 + 2x_2 + x_3 - m(a_1 + a_2)$

$Z + (-4m-3)x_1 + 9x_2 + (-m-1)x_3 + me_1 = -25m$

	x_1	x_2	x_3	a_1	a_2	e_1	e_2	sol	R
a_1	1	-1	1	1	0	0	0	15	15
a_2	3	1	0	0	1	-1	0	10	$10/3$
e_2	0	1	1	0	0	0	1	10	∞
Z	$4m-3$	9	$-m$	0	0	m	0	$-25m$	

a_1	0	$4/3$	1	1	1/3	0	$35/3$	$35/3$	
x_1	1	$1/3$	0	0	$-1/3$	0	$10/3$	∞	
e_2	0	1	1	0	0	1	10	10	
Z	0	$4m-3$	$-m$	0	$-m/3$	0	$-35m/3$	$-25m$	

a_1	0	$7/3$	0	1	$1/3$	-1	$5/3$	5	
x_1	1	$1/3$	0	0	$-1/3$	0	$10/3$	-10	
x_3	0	1	1	0	0	1	10	∞	
Z	0	$7m/3$	0	0	$-m/3$	m	$-5m/3$	$-25m$	

e_1	0	7	0	1	-3	5	$-5/7$		
x_1	1	-2	0	0	-1	5	$-5/2$		
x_3	0	1	1	0	0	1	10	10	
Z	0	7	0	0	-9	25			
e_1	0	0	7	1	4	75			
x_1	1	0	2	0	1	25			
x_2	0	1	1	0	1	10			
Z	0	0	7	0	5	95			

ALL $\dots \geq 0 \Rightarrow \text{opt. sol}$

Max $Z = 95$ $x_1 = 10$ $x_2 = 0$ $x_3 = 0$

II-3

2	1	3	10	$\Rightarrow$ ADD Dummy
4	5	6	20	
15	6	4	30	

a)

2	1	3	0
0	6	4	5
15	4	5	6

b)

2	1	3	0
0	6	4	5
15	4	5	6

All $\bar{c}_{ij} \leq 0 \Rightarrow \text{optimal solution}$

II-4

